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Theory and Methodology

Stochastic group preference modelling in the multiplicative AHP: A model of group consensus

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Abstract

This paper proposes a model for the representation of a consensus-seeking group decision making process under the multiplicative variant of the analytic hierarchy process (AHP). In particular, it proposes a group preference model which expresses the group's preference intensity judgements as random variables with associated probability distributions, and the structure of the multiplicative AHP is exploited to derive interval judgements for the alternatives' final impact scores as perceived by the group. These interval judgements can be used to determine the probability that a rank reversal may occur amongst alternatives due to the group preference uncertainty, i.e. to assess the stability of the final impact score vector, which in turn offers information about the degree of consensus within the group about the decision.

Keywords: Multiplicative analytic hierarchy process; Interval judgements of impact scores; Group preference model; Group consensus

1. Introduction and rationale

The analytic hierarchy process (AHP), originally developed by Saaty (1977, 1980), is a widely applied multicriteria decision making tool which utilises the concept of pairwise comparisons to arrive at a scoring and rank ordering of the alternatives under consideration. The decision maker (hereinafter referred to as the DM) provides a subjective cardinal judgement about the intensity of his preference for each alternative over each other alternative under each of a number of criteria or properties. This cardinal judgement is assumed to be a categorical descriptive judgement value taken from a set of such descriptive judgement values or categories and later during the analysis converted onto a numerical scale. The complete set of pairwise comparisons is synthesised into a scoring and rank ordering of the alternatives under consideration.

Barzilai et al. (1987), Barzilai and Golani (1991) and Barzilai (1992) observed that the AHP, since it is essentially based on ratio information, can be converted to a variant with a *multiplicative* structure. This form of the AHP is commonly referred to as the multiplicative AHP.

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Due to the simplicity of obtaining full DM input about his/her subjective feelings for all pertinent aspects of the decision problem, the transparency of the method to the DM, and the existence of a number of user-friendly software packages, the AHP and its variants have been successfully applied to many real-life decision problems. Subsequent extensions to aid in group decision making (see, for example, Basak and Saaty, 1989; Barzilai and Lootsma, 1994) have already been incorporated into the existing software.

A current group preference model within the multiplicative AHP (see Barzilai and Lootsma, 1994) includes the members of the group as a separate level of the hierarchy. That is, each member of the group effectively performs his/her own *separate* pairwise comparisons under each criterion, and the resulting impact scores are aggregated across the group members using a geometric mean aggregation and suitable weights for the group members. This leads to point estimates of the final impact scores and a rank ordering of the alternatives which might be attributable to the group *as a conglomerated whole*, but sheds no light on the degree of consensus reached by the group. In particular, often two or more leading alternatives end up with final aggregated impact scores which are very close to one another, and there is no indication as to whether the final top-ranking alternative is marginally preferred by *all* group members (in which case it is the *genuine* leading alternative, accepted by all) or whether there is serious discord within the group, with support being evenly spread amongst several leading alternatives (in which case there is *no genuine* single leading alternative, and hence there is unlikely to be unanimous support for the resulting decision, leading to later difficulties in implementation of the "preferred" decision). We believe it desirable to offer a measure of group consensus to help the group to appreciate the level of concordance of the decision taken.

The preference judgements of each of the DMs will, in general, not be identical to one another, even in a consensus-seeking group, i.e. there will be a degree of variation in their responses. If the group is looked upon as a single entity (a Global DM), the internal variation of responses may be viewed as the uncertainty of the group (or Global DM) about its response. If the preference judgements of the Global DM display elements of uncertainty, then the final impact scores of the alternatives are also uncertain and can best be represented by an *impact score interval* (rather than a single point). These impact score intervals might exhibit some degree of overlap, introducing uncertainty as to the true rank ordering of the alternatives attributable to the group as a whole (we call this *group rank uncertainty*). The family of rank reversal probabilities in the system caused by this group rank uncertainty offers an indication of the stability of the rank ordering of the alternatives. Note that this problem is unrelated to the well-documented controversy of the rank reversal phenomenon in the case of decision problems with single-valued paired judgements when an identical copy of an alternative is added to the problem (Dyer, 1990; Belton and Gear, 1983; Saaty, 1990; Schoner et al., 1992, 1993). In this paper the reversals of rank are caused by the group rank uncertainty only.

Uncertainty on the part of a single DM about his preferences in the AHP context was originally studied by Saaty and Vargas (1987) through a simulation approach. A uniformly distributed variate for each range of the DM's preference judgements was generated and used to compute the principal right eigenvector of the pairwise comparison matrices. Repeating this process a large number of times allowed the distribution of the components of the eigenvector to be empirically derived. Thereafter interval estimates for each component were set up, which were used to calculate the probabilities of reversal of ranks within the system of components. Saaty and Vargas' method, whilst tractable, appears somewhat cumbersome: there is a requirement to perform a complete simulation study each time uncertainty arises in a discrete multicriteria decision problem. Furthermore Stam and Silva (1994) criticize Saaty and Vargas (1987) on several counts. For instance (i) the method used to construct their impact score intervals is questioned (it is dependent on the level of confidence used), and (ii) the impact score interval for each component of the right principal eigenvector is computed independently of that for each other component, ignoring the possibility of correlation between components, which Stam and Silva show by way of an example to be not insignificant. Stam and Silva (1994) propose two further estimators of the probability of rank reversal, and offer a rigorous statistical analysis of the rank reversal likelihoods in any system.

Other research to incorporate uncertainty of pairwise judgements in the AHP were restricted to finite interval judgements (Arbel, 1989; Arbel and Vargas, 1993; Zahir, 1991; Salo and Hämäläinen, 1992). Arbel and Vargas (1993), propose an optimisation approach, in which the collection of the DM's algebraic preference statements results in a set of linear inequalities, constituting the constraints of a linear programming (LP) problem. Examination of the LP solution provides an indication of the robustness of the rank ordering of the alternatives to the uncertainty in the DM's preference judgements. Arbel and Vargas refer to their method as *preference programming*. Zahir (1991) also assumed uniform interval judgements by the DM; this approach was analytical and approximate to the first order for small dimension pairwise comparison matrices ($\leqslant 3$ alternatives), and numeric (but exact and computationally more complex) for larger dimensional problems. Zahir does not provide any statistical analysis of the rank reversal problem. Salo and Hämäläinen (1992) use preference programming to modify and fine-tune the DM's initially specified interval judgements in an attempt to minimise any inconsistency and ambiguity on the part of the DM, thus attempting to reduce the length of the judgement intervals, and hence increase the stability of the rank ordering. Preference programming was first proposed as a group decision support technique by Hämäläinen et al. (1991) in connection with an energy policy problem. Another attempt at modelling uncertainty or imprecision of responses in the AHP has used fuzzy logic (see Boender et al., 1989; Buckley, 1985; Van Laarhoven and Pedrycz, 1983).

In this paper we will examine variation in the DM's comparative judgements in a consensus-seeking group in the paradigm of the multiplicative AHP. We will exploit the mathematical structure of the method to determine the theoretical distribution of the alternatives' impact scores under variation in the preference judgements amongst the alternatives by the DMs. Thus interval judgements can be set up for the impact scores, and the probability of rank reversal amongst the alternatives caused by the group rank uncertainty can be calculated for the group's responses. This method will require only mild distributional assumptions about the variation in the subjective judgements across the DMs.

In Section 2 an overview of the multiplicative AHP is presented, followed by a derivation of the theoretical distribution of the impact scores when group members' preference judgements amongst alternatives are not all identical. This is followed by a brief review of two methods from the literature which might be used to measure the stability of the impact scores. A measure of the group's degree of consensus in a multicriteria decision is proposed and interpreted. Finally we illustrate our contribution by means of two examples: one consisting of hypothetical data, chosen to highlight features of the model, and the other being an actual real-world problem.

2. The multiplicative AHP for an individual decision maker: An overview

In the basic experiment, two stimuli S_j and S_k (two alternatives A_j and A_k under a particular criterion) are presented to the DM who is requested to express his graded comparative judgement about the pair. That is, he is asked to express his indifference between the two, or his weak, definite, strong or very strong preference for one of them over the other. We assume that stimuli have unknown subjective values V_j and V_k, and the purpose of the experiment is to approximate these values by the calculated impact scores. The DM's pairwise comparative judgement of S_j vis-à-vis S_k is captured on a *category scale* to restrict the range of possible verbal responses presented to the DM, and is characterised by an integer-valued index δ_{jk} designating the gradations of the DM's judgement according to the scale below.

Comparative judgement	Gradation index δ_{jk}
Very strong preference for S_k over S_j	−8
Strong preference for S_k over S_j	−6
Definite preference for S_k over S_j	−4
Weak preference for S_k over S_j	−2
Indifference between S_j and S_k	0
Weak preference for S_j over S_k	+2
Definite preference for S_j over S_k	+4
Strong preference for S_j over S_k	+6
Very strong preference for S_j over S_k	+8

Intermediate integer values can be assigned to δ_{jk} in order to express a hesitation between two adjacent gradations. In the multiplicative AHP the DM's judgement about the pair S_j and S_k is taken to be an estimate of the *preference ratio* V_j/V_k. The comparative judgements are converted into values on a geometric scale, characterised by scale parameter γ. Thus we define

$$r_{jk} = \exp(\gamma\delta_{jk})$$

to be the numeric estimate of the preference ratio V_j/V_k given by the DM. Although there is no unique scale of human judgement, a plausible value of γ is ln 2, implying a geometric scale with progression factor 2 (Lootsma, 1993).

Suppose that there are n decision alternatives under consideration. We will assume that the matrix of pairwise comparisons $\Delta = \{\delta_{jk}, j, k = 1, \ldots, n\}$ is skew-symmetric, i.e. $\delta_{kj} = -\delta_{jk}$. Thus a total of only $n(n-1)/2$ pairwise judgements are required to fully determine Δ. We can estimate the vector **V** of subjective stimulus values via logarithmic least squares regression (Lootsma, 1993). We thus approximate **V** by the vector **v** which minimises

$$\sum_{j<k} (\ln\ r_{jk} - \ln\ v_j + \ln\ v_k)^2. \tag{1}$$

Substituting $q_{jk} = \ln\ r_{jk} = \gamma\delta_{jk}$ and $w_j = \ln\ v_j$, the function to be minimised is

$$\sum_{j<k} (q_{jk} - w_j + w_k)^2, \tag{2}$$

as a function of the $w_j, j = 1, \ldots, n$. The associated set of normal equations can be written as

$$\gamma \sum_{k=1}^{n} \delta_{jk} = nw_j - \sum_{k=1}^{n} w_k, \quad j = 1, \ldots, n. \tag{3}$$

In principle, by solving (3) we obtain a solution w_j^* with an additive degree of freedom. Thereafter we calculate the vector $\mathbf{v}^*$ with components $v_j^* = \exp(w_j^*)$, and use the multiplicative degree of freedom in $\mathbf{v}^*$ to find a normalised solution $\mathbf{v}$. Note that the rank ordering of the components of v does not depend on the scale parameter γ (Lootsma, 1988, 1993). However there is *no unique solution* to the system in (3); by setting $\sum w_k = 0$ we obtain a particular solution. An unnormalised solution to the logarithmic least squares problem in (1) can then be explicitly written as

$$v_j = \exp(w_j) = \exp\left(\frac{1}{n}\gamma \sum_{k=1}^{n} \delta_{jk}\right) = \prod_{k=1}^{n} r_{jk}^{1/n}$$

i.e. the alternatives' impact scores v_j are found by the calculation of a *geometric mean* of the r_{jk} over all $k = 1, \ldots, n$. The v_j can, if desired, be *normalised* in the sense that $\sum v_j = 1$. This is merely cosmetic, however. In what follows we will work with the raw (unnormalised) impact scores.

Usually the choice of decision alternative has to be taken in the face of multiple conflicting criteria. The alternatives $A_j, j = 1, \ldots, n$ are thus compared pairwise with respect to each criterion $C_i, i = 1, \ldots, m$. Let r_{ijk} represent the numerical value on a geometric scale assigned to the DM's verbal estimate of V_{ij}/V_{ik}, the ratio of the subjective values of the alternatives under consideration under criterion C_i. Then $r_{ijk} = \exp(\gamma\delta_{ijk})$, where δ_{ijk} is the integer-valued index designating the DM's judgement. The impact scores v_{ij} of the alternatives $A_j, j = 1, \ldots, n$ under criterion $C_i, i = 1, \ldots, m$ are merely the geometric means of the rows of the pairwise comparison matrix under C_i. Final scores for the alternatives in the multiplicative AHP follow from the geometric mean aggregation rule: an unnormalised final score s_j for alternative A_j is computed as

$$s_j = \prod_{i=1}^{m} v_{ij}^{g_i} = \prod_{i=1}^{m} \left(\prod_{k=1}^{n} r_{ijk}^{1/n}\right)^{g_i} = \prod_{i=1}^{m}\prod_{k=1}^{n} \rho_{ijk}^{g_i}, \tag{4}$$

where

$$\rho_{ijk} = \exp\left(\frac{\gamma}{n}\delta_{ijk}\right),$$

and g_i is the normalised weight assigned to criterion C_i. Sometimes the criterion weights are determined by a pairwise comparison analysis amongst the criteria, analogous to the amongst alternatives under a single criterion. In this case the unnormalised criterion weights are calculated as the geometric means of the row entries in the pairwise comparison matrix of criterion preferences.

3. A stochastic group preference model within the multiplicative AHP

Assume that each individual DM in the group identifies a single value of δ_{ijk}, but that differences in subjective judgements lead to a *range* or interval of δ_{ijk} values across the members of the group. We will assume that the group members' propensity for choosing a value within this range can be modelled by some probability distribution, the variance of which is some measure of the lack of unanimity within the group about the comparison of alternatives j and k under criterion i. Thus δ_{ijk} *for the group* is a random variable, which in turn implies that $r_{ijk} = \exp(\gamma\delta_{ijk})$ is also a random variable. If we now let

$$\rho_{ijk} = \exp\left(\frac{\gamma}{n}\delta_{ijk}\right),$$

then the impact scores s_j in (4) are the product of a *sequence of random variables*. Alternatively we can write

$$\ln\, s_j = \sum_{i=1}^{m}\sum_{k=1}^{n} g_i \,\ln\, \rho_{ijk} = \frac{\gamma}{n}\sum_{i=1}^{m}\sum_{k=1}^{n} g_i\, \delta_{ijk},$$

since $\rho_{ijk} > 0$ always. We will assume that the group of DMs agree unanimously on the criterion weights (the g_i). (This would imply that there is absolute agreement about all the pairwise comparative judgements amongst criteria.) Alternatively the g_i have been set a priori by the DM or "problem owner". This assumption is appropriate in a wide range of applications. Under this premise the g_i can be considered constant. Then by a generalisation of the central limit theorem (Feller, 1971), ln s_j is approximately normally distributed, with the normal approximation improving as n and m get large, and as the distribution of ln ρ_{ijk} tends towards normality. The distribution of the overall impact scores s_j is thus asymptotically lognormal under these assumptions. In most practical applications we can expect m and n to be fairly small (both $m, n \leqslant 10$, say), but the product mn might be fairly large, implying a reasonable approximation of normality. Normality of ln s_j will thus not depend too severely on the distribution of the δ_{ijk}. Note that disparity between DMs is assumed to occur at a *single* level of the hierarchy only, i.e. in the preference judgements amongst alternatives. The distribution of the ln s_j is of unknown functional form if uncertainty occurs at more than one level of the hierarchy, and some other methodology [for example, a simulation approach (see Saaty and Vargas, 1987; Stam and Silva, 1994)] will have to be employed under these circumstances.

Let us assume now that δ_{ijk} is approximately normally distributed with mean $\mu_{\delta_{ijk}}$ and variance $\sigma^2_{\delta_{ijk}}$ (although these results will be approximately true for any reasonably symmetrical distribution for δ_{ijk}). This appears to be not an unreasonable assumption in a wide range of practical applications: in a group of DMs the value of δ_{ijk} is likely to be symmetrically distributed around some "most likely" value, with the probability of extreme values on either side of this being small. Under these circumstances ln s_j will be normally distributed with mean

$$\mu_{\ln\, s_j} = \frac{\gamma}{n}\sum_{i=1}^{m}\sum_{k=1}^{n} g_i\, \mu_{\delta_{ijk}} \tag{5}$$

and variance

$$\sigma^2_{\ln\, s_j} = \left(\frac{\gamma}{n}\right)^2\left[\sum_{i=1}^{m}\sum_{k=1}^{n} g_i^2\sigma^2_{\delta_{ijk}} + 2\sum_{i=1}^{m} g_i^2\left(\sum\sum_{k<p} \sigma_{\delta_{ijk},\delta_{ijp}}\right)\right], \tag{6}$$

where $\sigma_{\delta_{ijk},\delta_{ijp}}$ is the covariance measured between any pair of the variables under criterion $C_i, i = 1, \ldots, m$ for the given group of DMs. The general form of the variance term given here assumes that the components of the impact score vector are *not independent* of one another: dependence may arise from the DM's preference selections themselves. In practice the mean and variance of the distributions of each δ_{ijk} are easily calculated from the complete set of responses of all the individual DMs, as are the covariances between each pair of δ_{ijk} values under C_i.

We can now construct a set of interval judgements for the vector of overall impact scores. Then

$$s_j \in \exp\left[\mu_{\ln s_j} \pm z_{\alpha/2}\sqrt{\sigma^2_{\ln s_j}}\right], \quad j = 1, 2, \ldots, n$$

with $100(1-\alpha)\%$ certainty. Whilst interval judgements of the s_j are of interest in their own right, we seek to compute the probabilities that rankings amongst the alternatives may reverse due to the group rank uncertainty, which in turn will provide information as to the stability of the overall impact score vector. This will be based on the degree of overlap within the family of interval judgements for the s_j, following similar approaches to these of Saaty and Vargas (1987) and Stam and Silva (1994), which are briefly outlined and adapted for our use in Section 4. To this end we can more simply work with the symmetrical interval judgements of $\ln s_j$.

An important feature of the multiplicative AHP as a discrete multicriteria decision making tool is its simplicity. The group preference model proposed here obviously compromises this simplicity in favour of a more sophisticated model, both in terms of data and computational requirements. However we believe that the increased complexity is justified since the model offers assistance to a DM group in that it quantifies the effect and extent of the uncertainty of the group preferences that are not otherwise available. These are illustrated in the examples in Section 6.

4. Stability of the group's unnormalised impact score vector

We now outline two fundamentally different estimators to be found in the literature of the probability of rank reversal of the components of the impact score vector, and which have been adapted for our purposes.

4.1. The model of Saaty and Vargas (1987)

Construct a $(1-\alpha)$ level "interval of variation" for the logarithm of the ith component of the impact score vector (IV_i^α), choosing α sufficiently small. If the intersection of IV_i^α and $\mathrm{IV}_j^\alpha (i \neq j)$ is defined as IV_{ij}^α, then determine the estimate P_{ij} of the probability of rank reversal Π_{ij} associated with each pair of alternatives i and j by

$$P_{ij} = \begin{cases} 0 & \text{if } \mathrm{IV}_{ij}^\alpha = \phi, \\ \mathrm{P}(\ln s_j \in \mathrm{IV}_i^\alpha) & \text{if } \mathrm{IV}_i^\alpha \subseteq \mathrm{IV}_j^\alpha, \\ \mathrm{P}\left(\ln s_i \in \mathrm{IV}_j^\alpha\right) & \text{if } \mathrm{IV}_j^\alpha \subseteq \mathrm{IV}_i^\alpha, \\ \mathrm{P}\left(\ln s_i, \ln s_j \in \mathrm{IV}_{ij}^\alpha\right) & \left\{ \begin{array}{l} \text{if } \mathrm{IV}_{ij}^\alpha \neq \phi; \mathrm{IV}_i^\alpha, \\ \mathrm{IV}_j^\alpha \neq \mathrm{IV}_{ij}^\alpha. \end{array} \right. \end{cases}$$

Thus if IV_i^α and IV_j^α do not overlap then there is no uncertainty about the true rank ordering of i and j, and if they do overlap then P_{ij} is the probability that $\ln s_i$ and $\ln s_j$ are in the intersection IV_{ij}^α.

Problems with P_{ij} as an estimator are that IV_i^α depends on the probability level α (if α is made smaller, IV_{ij}^α automatically becomes larger), and that it implicitly assumes that the components of the impact score vector are statistically independent of one another, which is clearly not the case since the pairwise comparison matrices are known to be skew-symmetric. The extent of dependence will be based on the specific data at hand. However the correlations within the DMs' preference responses are indeed considered.

4.2. The model of Stam and Silva (1994) based on magnitude of preference

Assuming that rank reversal between two alternatives i and j occurs if alternative i would be preferred over j under perfect information, but is calculated to be less preferred based on the sample information on the interval judgements, then define an estimator P'_{ij} of Π_{ij} by examining the magnitude of the *difference* between the logarithms of the ith and jth component of the impact score vector, $D_{ij} = \ln s_i - \ln s_j$. Since the vector of logarithms of the impact scores has been shown in Section 3 to be multivariate normally distributed, D_{ij} is also normally distributed, with mean $\mu_{D_{ij}} = \mu_{\ln s_i} - \mu_{\ln s_j}$ and variance $\sigma^2_{D_{ij}} = \sigma^2_{\ln s_i} + \sigma^2_{\ln s_j} - 2\sigma_{\ln s_i, \ln s_j}$, where $\sigma_{\ln s_i, \ln s_j}$ is the covariance of the logarithm of the components of the impact score vector. This model thus includes the structural correlations between the components of the impact score vector arising from the skew-symmetry of the pairwise comparison matrices. Stam and Silva show that the estimate of P'_{ij} is given by $P'_{ij} = 2P(D_{ij} > 0)(1 - P(D_{ij} > 0))$. The probability $P(D_{ij} > 0)$ can be estimated using the maximum likelihood estimators $\bar{x}_{D_{ij}} = (\bar{x}_{\ln s_i} - \bar{x}_{\ln s_j})$ and $s^2_{D_{ij}}$ of $\mu_{D_{ij}}$ and $\sigma^2_{D_{ij}}$ respectively. By this definition P'_{ij} ranges from 0, when one alternative is always preferred to the other, to 0.5, when each alternative is equally likely to be preferred.

Stam and Silva's estimator implicitly includes both sources of correlation in its estimate of rank reversal probability. Since it is impossible in practice to determine the covariance of the logarithm of the components of the impact score vector for a decision making group by analytical means, they suggest to estimate these covariances by simulation from the original distributions of the response data. In a practical decision-making situation it is suggested that both methods described above (employing fundamentally different definitions of the probability of rank reversal) are used to obtain a good idea of an estimate of Π_{ij}.

Using either of the above estimators, the probability that alternative i will reverse rank with some other alternative due to the lack of unanimity in the group is then given by

$$P_i = 1 - \prod_{j=1}^{n}(1 - P_{ij}), \quad i = 1, 2, \ldots, n \tag{7}$$

and finally the probability of *at least one* rank reversal occurring in the system due to group preference uncertainty is given by

$$P = 1 - \prod_{1 \leqslant i \leqslant j \leqslant n} (1 - P_{ij}) \tag{8}$$

(replace P_{ij} by P'_{ij} in (7) and (8) when using Stam and Silva's model). The quantity P is a measure of the overall instability of the vector of unnormalised impact scores given the lack of unanimity in the group reflected in the range of responses for each pairwise comparison. However the group should examine not only P, but also P_j, $i = 1, \ldots, n$ and P_{ij} or P'_{ij}, $i, j = 1, \ldots, n$ before reaching a conclusion as to whether or not the lack of agreement in the system between DMs is too great to allow a reliable rank ordering of the alternatives for the purpose at hand. Often the objective of a multicriteria analysis is to identify and rank order just a small number of the top-ranking alternatives (say i out of the n, $i < n$). All that is important in this case is the probability of rank reversal amongst the top-ranking i alternatives (and possibly the next-ranking one or two): rank reversals amongst the other alternatives are irrelevant.

5. Interpretation of the group preference model

The methodology developed here offers an alternative representation of the group preference process for a consensus-seeking group. Instead of each DM performing his own separate set of pairwise comparisons (i.e. the DMs constituting a separate level of the hierarchy), this model has the group working *together* to derive a distribution for each of the comparative judgements for the group (the δ_{ijk}). The variance of these distributions indicates the degree of discord within the group about a particular pairwise comparison. After aggregation, the interval judgements of the final impact scores of the alternatives can be used to synthesise an indication of the overall degree of consensus within the group about the rank ordering and final impact scores of the alternatives. In particular, we propose that the group's overall *degree of consensus*, Γ, be computed as the probability of no possible rank reversal in the system due to the lack of unanimity in the group's responses, i.e. $\Gamma = 1 - P$, where P is the probability of at least one rank reversal in the system caused by disagreement amongst the group members. Thus $0 \leqslant \Gamma \leqslant 1$. Clearly it is desirable that Γ be close to 1 for the group to declare sufficient consensus having been reached amongst the members, but exactly what level of Γ is reasonable as a cut-off point to indicate a "high" degree of consensus is, of course, entirely subjective, and depends on the composition of the group and the purpose, importance and repercussions of the decision. A low level of Γ would almost certainly lead to discontent within the group about the decision, however consensus-seeking its composition is, and may jeopardise successful implementation of the decision.

6. Illustrative examples

6.1. An illustration of the methodology

We first illustrate the effect of lack of unanimity in preference judgements between DMs by using an example of $n = 4$ alternatives and $m = 2$ criteria within a group of four DMs. The complete response data is given in Appendix A, whilst the pairwise comparison matrices of δ_{1jk} and δ_{2jk} values are given in Table 1, where each cell entry represents $(\mu_{\delta_{ijk}}; \sigma^2_{\delta_{ijk}})$, the mean and variance of the normal distribution describing the group's range of preference judgements, δ_{ijk}. We will assume that the pairwise comparison matrices are skew-symmetric, i.e. that $\mu_{\delta_{ijk}} = -\mu_{\delta_{ikj}}$ and $\sigma^2_{\delta_{ijk}} = \sigma^2_{\delta_{ikj}}$. Furthermore, $\sigma^2_{\delta_{ijk}} = 0$ implies that there is complete agreement in estimating δ_{ijk}. We will assume throughout that $\gamma = \ln 2$, and that the criteria have equal weights, i.e. $g_i = 0.5$, $i = 1, 2$. To facilitate later comparison we now compute final scores for the alternatives under a scenario of *complete agreement* within the group (which is equivalent to assuming that $\sigma^2_{\delta_{ijk}} = 0$ for all i, j, k, and is the result using the existing model of group preference aggregation). Then under the multiplicative AHP, normalised final scores are $s_1 = 0.4062$, $s_2 = 0.2415$, $s_3 = 0.3132$ and $s_4 = 0.0391$. Thus A_1 appears to be a distinct first choice, with A_3 followed by A_2 a clear distance behind. Consideration of these results would leave a decision making group with little hesitation as to the choice of leading alternative. Now consider the effect of the *lack of unanimity* within the group as indicated in Table 1 on the final scores.

Table 1
Pairwise comparison matrices

	A_1	A_2	A_3	A_4
δ_{1jk}				
A_1	(0;0)	(2; 0.67)	(2; 3.33)	(4; 0.67)
A_2		(0; 0)	(2; 1.33)	(2; 4)
A_3			(0; 0)	(2; 0)
A_4				(0; 0)
δ_{2jk}				
A_1	(0; 0)	(1; 4.67)	(−2; 0.67)	(2; 2.67)
A_2		(0; 0)	(−1; 0.67)	(3; 2)
A_3			(0; 0)	(5; 2)
A_4				(0; 0)

Then from (5) and (6):

$$\mu_{\ln s_1} = \frac{\ln 2}{4}[0.5(0 + 2 + 2 + 4) + 0.5(0 + 1 - 2 + 2)] = 0.7798,$$

$$\begin{aligned}\sigma^2_{\ln s_1} = {} & \left(\frac{\ln 2}{4}\right)^2 0.5^2[(0 + 0.67 + 3.33 + 0.67) \\ & + (0 + 4.67 + 0.67 + 2.67) \\ & + 2((0 + 0 + 0 + 0.67 + 0.67 + 0.67) \\ & + (0 + 0 + 0 - 1 + 3.33 - 0.67))] \\ = {} & 0.1501.\end{aligned}$$

Thus $\ln s_1 \sim N(0.7798; 0.1501)$. Invoking the Saaty and Vargas model, the interval of variation $\mathrm{IV}_1^{0.1}$ is [0.1444; 1.4152] with 90% certainty. The means and standard deviations of $\ln s_j$, $j = 1, 2, 3, 4$ are given in Table 2.

Intervals of variation for $\ln s_j$, $j = 2, 3, 4$ with $\alpha = 0.1$ are thus:

$\ln s_2 \varepsilon$ [−0.5096; 1.0294],

$\ln s_3 \varepsilon$ [−0.1364; 1.1762],

$\ln s_4 \varepsilon$ [−1.8438; −1.2754].

Clearly the intervals for A_1, A_2 and A_3 exhibit a large degree of overlap, i.e. there is a significant

Table 2
Means and standard deviations of $\ln s_j, j = 1, 2, 3, 4$

j	$\mu_{\ln s_j}$	$\sigma_{\ln s_j}$
1	0.7798	0.3874
2	0.2599	0.4692
3	0.5199	0.4002
4	−1.5596	0.1733

chance that rank reversal could occur between these three alternatives, using the Saaty and Vargas definition. The decision as to the subjective group rank ordering of the alternatives thus becomes much less clear under this model. This is caused to a great degree by large correlations present in the DMs' responses. Furthermore, note that the final impact scores under conditions of complete agreement amongst group members correspond to the normalised means of the s_j values derived from Table 2. That is $s_j = \exp(\mu_{\ln s_j})$. Thus the multiplicative AHP under complete agreement is merely a special case of this more general group preference model.

The probabilities that a rank reversal could occur between any pair of the alternatives are now calculated according to the method outlined in Section 4.1:

$$P_{12} = P(0.1444 \leqslant \ln s_1 \leqslant 1.0294) \\ P(0.1444 \leqslant \ln s_2 \leqslant 1.0294) \\ = (0.6903)(0.5472) = 0.3777,$$

$$P_{13} = P(0.1444 \leqslant \ln s_1 \leqslant 1.1762) \\ P(0.1444 \leqslant \ln s_3 \leqslant 1.1762) \\ = (0.7969)(0.7760) = 0.6184,$$

$$P_{14} = 0,$$

$$P_{23} = P(-0.1364 \leqslant \ln s_2 \leqslant 1.0294) \\ P(-0.1364 \leqslant \ln s_3 \leqslant 1.0294) \\ = (0.7508)(0.8485) = 0.6371,$$

$$P_{24} = 0, \quad P_{34} = 0.$$

Thus the probabilities $P_i, i = 1,2,3,4$ that alternative i will reverse rank with another alternative follow as:

$$P_1 = 1 - (1 - P_{12})(1 - P_{13})(1 - P_{14}) \\ = 1 - (0.6223)(0.3816)(1) = 0.7625,$$
$$P_2 = 1 - (1 - P_{12})(1 - P_{23})(1 - P_{24}) \\ = 1 - (0.6223)(0.3629)(1) = 0.7742,$$
$$P_3 = 1 - (1 - P_{13})(1 - P_{23})(1 - P_{34}) \\ = 1 - (0.3816)(0.3629)(1) = 0.8615,$$
$$P_4 = 1 - (1 - P_{14})(1 - P_{24})(1 - P_{34}) = 0.$$

Finally, the probability of at least one rank reversal in the system is

$$P = 1 - (1 - P_{12})(1 - P_{13})(1 - P_{14})(1 - P_{23}) \\ (1 - P_{24})(1 - P_{34}) = 0.9138$$

i.e. a very large chance that the lack of agreement amongst group members could lead to an interchange of ranking of at least two of the alternatives. Conversely put, the group's overall degree of consensus, Γ, is a low 0.0862, casting doubt on the implementability of simply choosing alternative A_1 as a cut-and-dried "best" alternative for the group. Table 3 shows the scoring and rank ordering for each of the individual DMs.

It is evident that DMs 1, 2 and 3 agree fully on a rank ordering of the alternatives, but DM 4 differs considerably on this, especially in the choice of the leading alternative. This large difference in the scoring of the alternatives amongst DMs lead to the wide intervals of variation observed above. As already mentioned, the probabilities produced by this method depend on the value of α used; repeating the example with $\alpha = 0.01$ yields the values of P_i and P as $P_1 = 0.9895$, $P_2 = 0.9890$, $P_2 = 0.9956$, $P_4 = 0$ and $P = 0.9993$. There is obviously no "correct" value for α; however it should not be so small as to include too much of the tail probability of the intervals of variation. Experience suggests a value of α of the order of 0.1 or 0.2 should be satisfactory.

We now invoke the Stam and Silva model. A simulation experiment was carried out in which 100 pairwise comparison matrices were created, using the distributional data given in Table 1 as source. Values for $\ln s_j, j = 1, 2, 3, 4$ were calculated from these matrices. Relevant summary statistics are presented in Tables 4 and 5.

Table 3
Scoring and rank ordering (in parentheses) for four DMs

j	DM1	DM2	DM3	DM4
1	0.414 (1)	0.525 (1)	0.448 (1)	0.218 (2)
2	0.207 (3)	0.120 (3)	0.205 (3)	0.518 (1)
3	0.348 (2)	0.312 (2)	0.317 (2)	0.218 (2)
4	0.031 (4)	0.043 (4)	0.031 (4)	0.046 (4)

Table 4
Summary statistics from the simulation study

j	$\bar{x}_{\ln s_j}$	$s^2_{\ln s_j}$
1	0.7848	0.1104
2	0.2725	0.1385
3	0.4996	0.6716
4	−1.5569	0.1196

Table 5
Correlation matrix from the simulation study

	$\ln s_2$	$\ln s_3$	$\ln s_4$
$\ln s_1$	−0.409	−0.331	−0.272
$\ln s_2$		−0.198	−0.534
$\ln s_3$			−0.218

From this data the mean and variance of the difference between the logarithms of the ith and jth components of the impact score vector ($\mu_{D_{ij}}$ and $\sigma^2_{D_{ij}}$) can be calculated, as in Table 6, and hence so can $P(D_{ij} > 0)$ and P'_{ij}.

Then $P_1 = 1 - (1 - 0.3119)(1 - 0.4014) = 0.5881$ and likewise $P_2 = 0.6127$; $P_3 = 0.6631$, $P_4 = 0.0036$, which finally yields $P = 1 - (0.6881)(0.5986)(0.5628)(0.9964) = 0.7690$ (or alternatively $\Gamma = 0.2310$).

Table 7 shows the frequency of occurrence of each rank ordering in the simulation experiment. The rank ordering (1,3,2,4) was most frequently observed: This is the outcome observed under the existing model of group preference aggregation. However in 33% of the simulations alternative 1 was *not* the leading alternative.

Table 6
Summary statistics on the differences, D_{ij}

i	j	$\mu_{D_{ij}}$	$\sigma^2_{D_{ij}}$	$P(D_{ij} > 0)$	P'_{ij}
1	2	0.5123	0.3501	0.8067	0.3119
1	3	0.2852	0.2346	0.7220	0.4014
1	4	2.3417	0.2926	1	0
2	3	−0.2271	0.2439	0.3228	0.4372
2	4	1.8295	0.3956	0.9982	0.0036
3	4	2.0566	0.2259	1	0

Table 7
Frequency of occurrence of each rank ordering

Rank ordering	Frequency (%)
(1, 2, 3, 4)	18
(1, 3, 2, 4)	49
(2, 1, 3, 4)	7
(2, 3, 1, 4)	5
(3, 1, 2, 4)	15
(3, 2, 1, 4)	6
Other rank orderings	0

The numerical results of the two methods differ considerably from one another since they are based on entirely different estimators of the probability that a rank reversal may occur. However the $P_{ij}(')$ and P_i values exhibit identical pattern of ordering in both models. In both cases the group's degree of consensus in this example is shown to be low, and clearly the members will have to review their individual preference intensities in an effort to reduce the rank reversal probabilities and increase consensus to ensure an implementable decision.

6.2. A case study: Strategy planning for dam water release management

The Kruger National Park is situated in the extreme North–East of South Africa, on that country's border with Mozambique. Several rivers traverse the Park from the high-lying land to the west of the Park towards the Mozambique coast. These rivers are an important source of irrigation water for farmers upstream of the Park, as well as supplying domestic and irrigation water to small rural settlements in the vicinity of the river. Because of the seasonal nature of rivers in this area and the unpredictability of the rainfall, the Department of Water Affairs and Forestry (DWAF) in South Africa embarked on a plan to build a dam on the Sabie River a short distance upstream of the western boundary of the Kruger National Park to ensure that sufficient water is available to users all the year round. Whilst seasonality in the flow in the rivers may be a negative factor for farmers and the surrounding settlements, it plays an important role in preserving the natural condition of the rivers' ecosystems. The effect of the construction of a dam upstream of the Park would naturally result in altered water flow patterns, and hence possible damage to ecosystems in the Park.

The Kruger National Park Rivers Research Group (KNPRRG) is a group consisting of environmental scientists and freshwater ecologists whose main aim is to ensure that the ecology of the rivers in the Park remains in as pristine a condition as possible for the benefit of the environment as a whole. As an exercise to test the suitability of using interactive MCDM techniques as a decision aid in integrated environmental management, the KNPRRG organised a facilitated workshop with the aim of planning a strategy for the management of water release from the new dam. A range of feasible scenarios were put forward for consideration, variations of which could form the basis of the group's preferred future strategy.

Strategy 1: Retain as much water in the dam as possible, allowing the maximum possible abstraction for irrigation by farmers. Release water from the dam only when the dam is full.

Strategy 2: Guarantee a minimum *dry season* flow of 1 m^3/s in the Sabie River downstream of the dam. Natural overflow will be allowed in the rainy season.

Strategy 3: Allow a flow of 30% of the natural (pre-development) flow in the Sabie River downstream of the dam all the year round.

Strategy 4: Allow present-day flows to continue, except in times of low rainfall or drought, in which case water would be retained in the dam.

Scenario 1 is clearly the most undesirable from the point of view of the KNPRRG, as it deviates most from the natural flow patterns of the river. On the other hand dam management would find it the easiest to implement, since all the other strategies would require continual active management.

Initial discussions revolved around choice of criteria. After some debate it was agreed that scenarios be judged under the effects of the *quantity* of water flow that would have naturally occurred in the absence of the dam, and the effects on water *quality*. The performance of a given strategy under the issue of quantity of water released was felt to depend on whether the flows would have been high, intermediate or low. Thus the problem was devolved into one of four alternatives (the scenarios) to be evaluated under four criteria (high, intermediate and low flows, and water quality). Agreed normalised criterion weights were as follows:

Effects at high flows	0.22
Effects at intermediate flows	0.30
Effects at low flows	0.37
Effects on water quality	0.11

The data collected at the workshop were later converted into gradation index values, and parameters of normal distributions for the group for each gradation index under each of the criteria estimated. The differences in preference intensity across members of the DM group were reflected as the variances of the distributions. Means and standard deviations for the logarithm of the final score for each of the four alternative scenarios are shown in Table 8, as are group intervals of variation (with $\alpha = 0.1$) for each $\ln s_j$.

If the facilitator were simply to take the midpoint (or mean) of the group's range of responses for each pairwise comparison as the group's *compromised* response, there would be no doubt as to the rank ordering of the scenarios: scenario 4 would be the first choice, followed by scenario 3, scenario 2, and finally scenario 1 as the least preferred option. Incorporating the present model of group preference aggregation, it is clear from Table 8 that scenario 4 is still far and away the group's preferred option. The rank ordering of the other alternatives is, however, by no means cut and dried: the intervals for scenarios 1 and 2 overlap to a large extent, and the upper endpoint for the interval for scenario 2 almost adjoins the

Table 8
Means, standard deviations and group intervals of variation ($\alpha = 0.1$) for $\ln S_j$ for the four scenarios

j	$\mu_{\ln s_j}$	$\sigma_{\ln s_j}$	Interval of variation
1	−1.2546	0.3290	(−1.7942; −0.7150)
2	−0.8682	0.3624	(−1.4625; −0.2739)
3	0.2235	0.3152	(−0.2934; 0.7404)
4	1.8992	0.1944	(1.5804; 2.2180)

Table 9
Summary statistics from the simulation for the KNPRRG decision

i	j	$\mu_{D_{ij}}$	$\sigma^2_{D_{ij}}$	$P(D_{ij} > 0)$	P'_{ij}
1	2	−0.3498	0.2426	0.7612	0.3635
1	3	−1.4791	0.4065	0.9898	0.0202
1	4	−3.1187	0.1524	1	0
2	3	−1.1293	0.4580	0.9524	0.0907
2	4	−2.7689	0.2419	1	0
3	4	−1.6396	0.3802	0.9961	0.0078

lower endpoint of the interval for scenario 3 for this level of α. Using the Saaty and Vargas definition of rank reversal we obtain $P_{ij} = 0\ \forall_{ij}$ except $P_{12} = 0.4212$ for $\alpha = 0.1$. Thus $P_i = 0.4212$ for $i = 1,2$ and $P_i = 0$ for $i = 3,4$ and $P = 0.4212$. Employing the Stam and Silva model, a simulation using the original distributions of pairwise judgements as source was run which yielded the results found in Table 9.

Using this definition of rank reversal yields $P_1 = 0.3764$, $P_2 = 0.4212$, $P_3 = 0.1160$ and $P_4 = 0.0078$, and $P = 0.4373$. This approach indicates that other reversals of rank between the various alternatives may occur, although these reversals all have small probability except for the case of scenarios 1 and 2. These results are generally in keeping with those generated by the Saaty and Vargas model. The actual decision reached by the KNPRRG group (i.e. select scenario 4) was not affected by superimposing this stochastic group decision model upon their workshopped results ($P_4 \approx 0$ using both models). However the model indicates that there was a fair degree of dissent within the group about the respective rankings of, particularly, scenarios 1 and 2; if a group ranking of *all* the scenarios had been required, this may have been more troublesome for the group to agree upon (the degree of group consensus, Γ, was only ≈ 0.57).

7. Conclusions

The lack of complete consensus within a decision-making group about preference intensities can lead to a final rank ordering of the alternatives which might appear to be non-optimal for a significant proportion of the group. We have exploited the structure of the multiplicative AHP to derive group interval judgements of the geometric row means of the pairwise comparison matrices and of the final unnormalised impact scores, which in turn allows the group to measure the degree of consensus present in their decision as represented by the final impact scores. This methodology is a generalisation of the standard multiplicative AHP for individual decision makers; indeed the multiplicative AHP for individual DMs is merely a special case when all variances equal zero.

This interpretation of the group decision process highlights the differences between the individual DM's responses. The result is that we should take care before making statements such as "A_1 has a large impact score than A_2 and A_3", which would be the interpretation of the group's decision under existing group aggregation models within the AHP. The clear-cut rank ordering can take on a very indistinct nature under lack of unanimous pairwise judgements. The group should consider the rank reversal probabilities before reaching a conclusion; individual perceptions of importance and risk aversion play a significant role, and we make no attempt to prescribe what constitutes a "best" alternative.

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Appendix A. Raw data used in the illustrative example (Section 6.1)

The following matrices represent the ordered preference responses δ_{ijk} by four DMs about $j = 4$ alternatives under $i = 2$ criteria.

	A_2	A_3	A_4
δ_{1jk}			
A_1	2 3 2 1	4 3 0 1	4 5 4 3
A_2		1 1 3 3	3 −1 3 3
A_3			2 2 2 2
δ_{2jk}			
A_1	1 2 3 −2	−3 −2 −2 −1	2 2 4 0
A_2		−2 −1 −1 0	3 2 2 5
A_3			6 6 5 3

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